Project Initialization and Planning Phase

Date	07-07-2026
Team ID	
Project Title	Smart Agricultural Production Optimization Engine (OptiCrop)
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) Report

This proposal report describes OptiCrop, a machine learning-based crop recommendation system that helps farmers choose the most suitable crop for their soil and climate conditions. The system analyzes 7 environmental parameters and predicts the best-fit crop from 22 possible options, aiming to reduce guesswork in crop selection and support more informed, data-driven farming decisions.

Project Overview	
Objective	The primary objective is to help farmers select the most suitable crop for their land by analyzing Nitrogen, Phosphorus, Potassium, temperature, humidity, pH, and rainfall values using a trained machine learning model, replacing guesswork with a data-backed recommendation.
Scope	The project covers 22 crop types represented in the training dataset (e.g. rice, maize, cotton, coffee, various fruits, pulses, and lentils). It provides a web-based interface accessible from any browser, with no login or installation required. The current scope does not include regional language support, voice input, SMS alerts, or a mobile app — these are noted as possible future extensions, not part of the current build.

Problem Statement	
Description	Farmers often choose crops based on tradition or advice from neighbors rather than actual soil and climate data, which can lead to poor yields and inefficient use of fertilizer and water. Scientific soil testing is often inaccessible or costly for small and marginal farmers, leaving them without a reliable way to match their land's conditions to the crop most likely to succeed. Unpredictable weather adds further risk, and there is no single, easy-to-use tool that combines soil health and climate data into one clear recommendation.
Impact	By providing an evidence-based crop recommendation, OptiCrop aims to help farmers make more confident planting decisions, potentially reducing the risk of choosing a poorly-suited crop and supporting more efficient use of fertilizer and water resources. As a student/portfolio-scale project, these are intended outcomes rather than measured, verified results from real-world deployment.

Proposed Solution	
Approach	The system uses a Logistic Regression model, trained on a balanced dataset of 2,200 samples across 22 crop classes, with features scaled using StandardScaler prior to training. K-Means Clustering is additionally applied for exploratory analysis of crop-environment groupings, useful for research or policy-oriented insight. The trained model is served through a Flask web application, deployed on Render.
Key Features	OptiCrop Prediction Engine: recommends the best-fit crop with 97.3% test accuracy using soil (N, P, K) and climate (temperature, humidity, pH, rainfall) data. Multi-page Web Interface: Home, About, and Find Your Crop pages with a styled, farmer-friendly design. Instant Results: prediction returned immediately after form submission, no login required. Exploratory Clustering: K-Means-based grouping of crops by environmental profile, useful for research-oriented analysis.

Resource Requirements

Resource Type	Description	Specification / Allocation
Hardware		
Computing Resources	Standard development machine — no GPU required, since Logistic Regression and K-Means are lightweight, CPU-based algorithms	Local: Intel Core i3 or above, 4 GB RAM minimum. Hosting: Render free tier (512 MB RAM, shared CPU)
Memory	RAM for model training and Flask application runtime	Training: under 1 GB RAM (small dataset). Production: well within Render's 512 MB free-tier allocation
Storage	Disk space for dataset, trained model, and source code	Dataset (CSV): under 200 KB. Trained model + scaler (.pkl files): a few hundred KB combined. Total repository size: well under 50 MB
Software		
Frameworks	Python web and ML frameworks	Flask — lightweight web framework for the prediction interface. Scikit-learn — Logistic Regression, K-Means, StandardScaler
Libraries	Data science and analysis libraries	pandas, numpy — data preprocessing. matplotlib, seaborn — EDA visualizations. pickle — model persistence
Development Environment	IDE and tools	Jupyter Notebook — model development and EDA. VS Code — Flask application development. Git & GitHub — version control
Data		
Dataset Source	Crop Recommendation Dataset (Kaggle — Smart Agricultural Production Optimizing Engine)	2,200 samples, 22 crop classes (100 each), 7 features (N, P, K, temperature, humidity, pH, rainfall), CSV format, under 200 KB total

Note: this proposal intentionally scopes the project to what has been implemented and tested — a Flask-based web application using classical ML models on a public dataset, deployed on a free-tier host. It does not claim enterprise-scale infrastructure (GPU training, auto-scaling, government dashboards) or unverified impact statistics, since none of that reflects the actual current build.